Interpretable Machine Learning

Permutation Feature Importance (PFI)

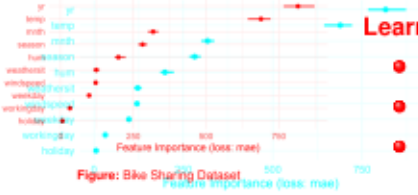

Figure: Bike Sharing Dataset

Learning goals

- Understand how PFI is computed
- Understanding strengths and weaknesses
- Testing Importance

PERMUTATION FEATURE IMPORTANCE (PFI)

Breiman (2001)

Idea: "Destroy" feat. of interest x_j by perturbing it s.t. it becomes uninformative, e.g., randomly permute obs in x (marginal distribution $\mathbb{P}(x_j)$ stays the same).

PFI for features x_S using test data $\mathcal{D}$:

- Measure the error **without permuting feat.** and **with permuted feat. values** $\tilde{x}_S$
- Repeat permuting the feat. (e.g., m times) and avg. the difference of both errors:

$$\widehat{PFI}_S = \frac{1}{m}\sum_{k=1}^{m} \mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}^S_{(k)}) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D}),$$

where $\mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D}) = \frac{1}{n}\sum_{(x,y)\in\mathcal{D}} L(\hat{f}(x), y)$

Interpretable Machine Learning – 1 / 12

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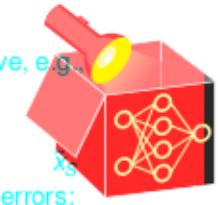

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The data $\mathcal{D}$ where x_S is replaced with $\tilde{x}^S$ is denoted as $\tilde{\mathcal{D}}^S$.

Example of permuting feature x_S with $S = \{1\}$ and $m = 6$:

Note: The S in x_S refers to a **S**ubset of features for which we are interested in their effect on the prediction.

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Interpretable Machine Learning – 1 / 12

PERMUTATION FEATURE IMPORTANCE

$\tilde{\mathcal{D}}^S_{(k)}$ $\mathcal{D}$ $\mathcal{D}$

i	x_S	x_1	x_2	x_3	x_1	x_2	x_3	x_1	x_2	x_3
1	2	4	7	4	1	4	7	1	4	7
⋮	1	5	8	5	2	5	8	2	5	8
n	3	6	9	6	3	6	9	3	6	9

1. **Perturbation:** Sample feature values from the distribution of x_S ($P(X_S)$).
   ⇒ Randomly permute feature x_S
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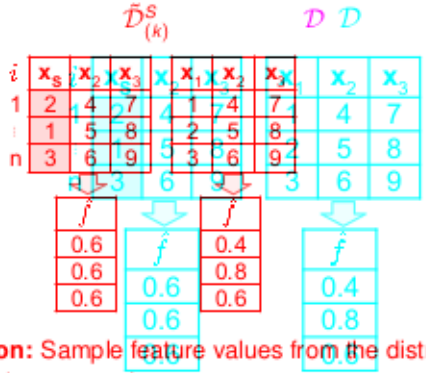
Interpretable Machine Learning – 2 / 12

PERMUTATION FEATURE IMPORTANCE



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Interpretable Machine Learning – 2 / 12

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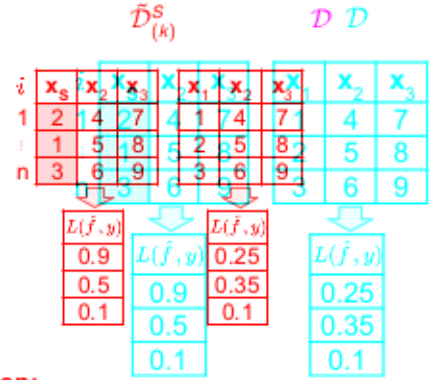

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Interpretable Machine Learning – 2 / 12

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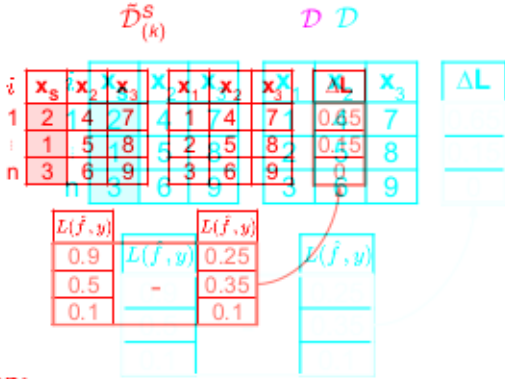

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Interpretable Machine Learning – 2 / 12

PERMUTATION FEATURE IMPORTANCE

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i	x_s	x_2	x_3	x_1	x_2	x_3	ΔL	x_3	ΔL
1	2	4	7	1	4	7	0.65	7	0.65
i	1	5	8	2	5	8	0.15	8	0.675
n	3	6	9	3	6	9	0	9	0

= 0.267

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Interpretable Machine Learning – 2 / 12

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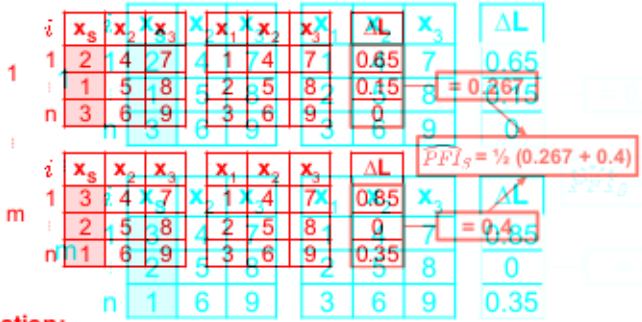

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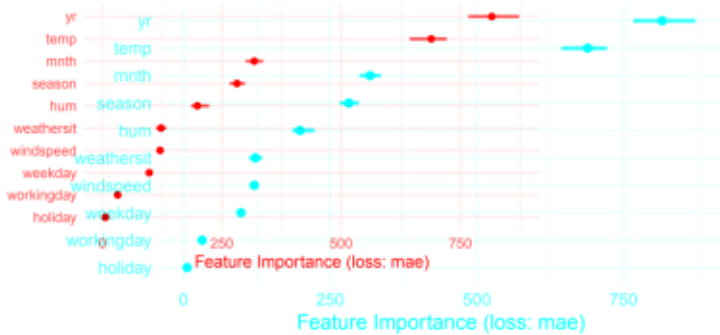
Interpretable Machine Learning – 2 / 12

EXAMPLE: BIKE SHARING DATASET



Interpretation:

- Year (yr) and Temperature (temp) are most important features
- Destroying information about yr by permuting it increases mean absolute error of model by 816
- 5% and 95% quantile of repetitions due multiple permutations are shown as error bars

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Interpretable Machine Learning – 3 / 12

COMMENTS ON PFI

- Interpretation: PFI is the increase of model error when feature's information is destroyed

Interpretable Machine Learning – 4 / 12

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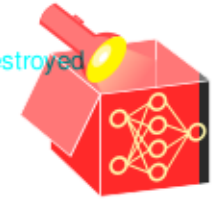

Interpretable Machine Learning – 4 / 12

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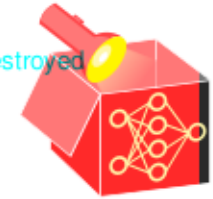

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  - PFI automatically includes importance of interaction effects with other features
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    - ⇒ Importance of all interactions with the permuted feature are contained in PFI score

Interpretable Machine Learning – 4 / 12

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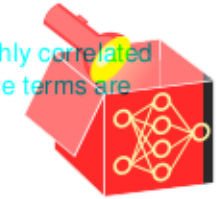
COMMENTS ON PFI - EXTRAPOLATION

Example: Let $y = x_3 + \epsilon_y$ with $\epsilon_y \sim N(0, 0.1)$ where $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$ are ϵ_2 are highly correlated highly correlated ($\epsilon_1 \sim N(0, 1)$, $\epsilon_2 \sim N(0, 0.01)$) and $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim N(0, 1)$. All noise terms are independent. Fitting a LM yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$.



Interpretable Machine Learning – 5 / 12

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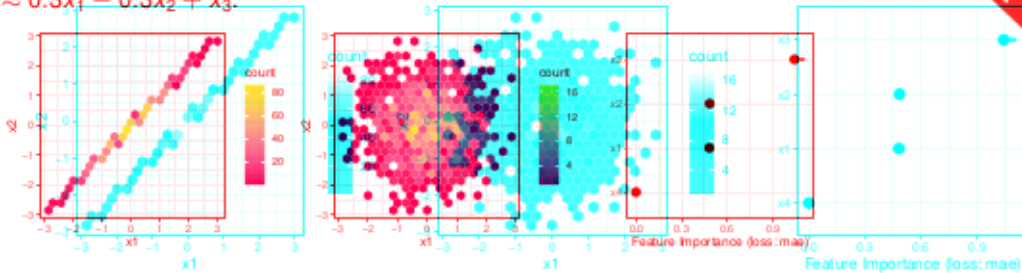

Hexbin plot of x_1, x_2 before permuting x_1 (left), after permuting x_1 (center), and PFI scores (right)

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Interpretable Machine Learning – 5 / 12

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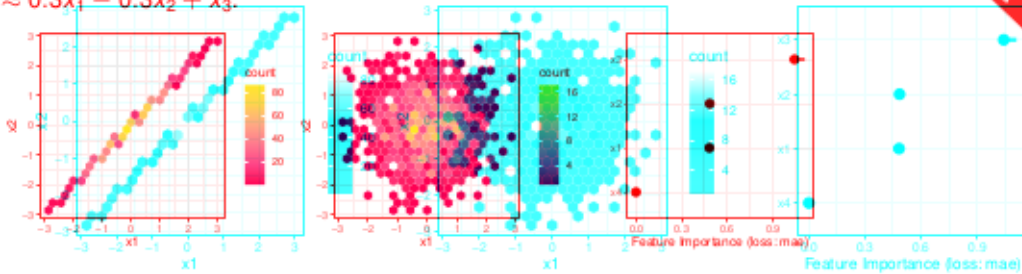

Hexbin plot of x_1, x_2 before permuting x_1 (left), after permuting x_1 (center), and PFI scores (right) $\Rightarrow$ x_1 and x_2 should be irrelevant for the prediction $\hat{f}(\mathbf{x})$ for $\{\mathbf{x} : \mathbb{P}(\mathbf{x}) > 0\}$ as $0.3x_1 - 0.3x_2 \approx 0$

$\Rightarrow$ PFI evaluates model on unrealistic obs. outside $\mathbb{P}(\mathbf{x})$ $\rightsquigarrow$ x_1, x_2 are considered relevant (PFI > 0)

COMMENTS ON PFI - INTERACTIONS

Example: Let $x_1, x_2, \ldots, x_4$ be independently and uniformly sampled from $\{-1, 1\}$ and

$$y := x_1 x_2 + x_3 + \epsilon_Y \text{ with } \epsilon_Y \sim N(0, 1)$$

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Interpretable Machine Learning – 6 / 12

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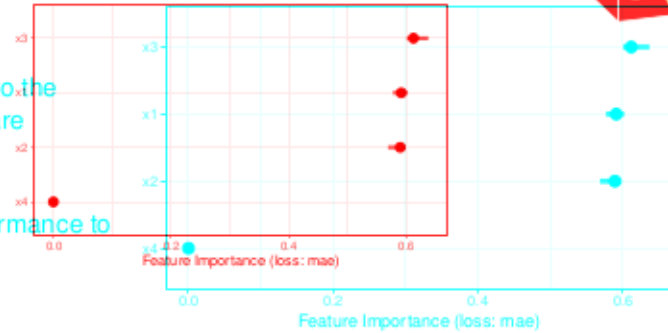
Although x_3 alone contributes as much to the prediction as x_1 and x_2 jointly, all three are considered equally relevant.

$\Rightarrow$ PFI does not fairly attribute the performance to the individual features.



Interpretable Machine Learning – 6 / 12

COMMENTS ON PFI - TEST VS. TRAINING DATA

Example: $x_1, \ldots, x_{20}, y$ are independently sampled from $\mathcal{U}(-10, 10)$. An `xgboost` model with default hyperparameters is fit on a small training set of 50 observations. The model overfits heavily.

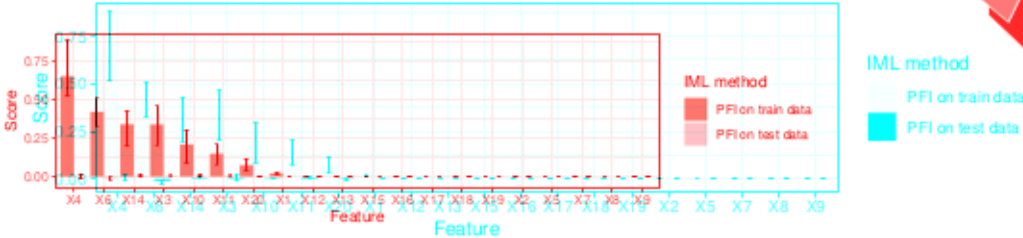


Figure: While PFI on test data considers all features to be irrelevant, PFI on train data exposes the features on which the model overfitted.

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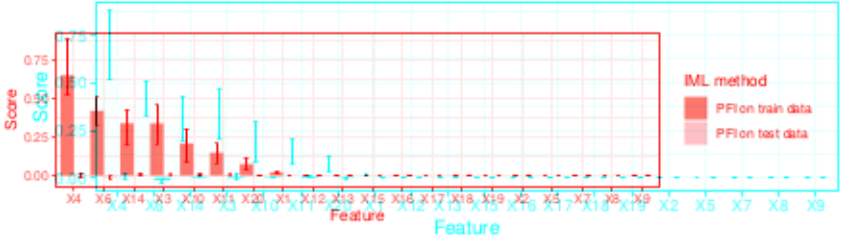

Figure: While PFI on test data considers all features to be irrelevant, PFI on train data exposes the features on which the model overfitted.

Why? PFI can only be nonzero if the permutation breaks a dependence in the data. Spurious correlations help the model perform well on train data but are not present in the test data.

⇒ If you are interested in which features help the model to generalize, apply PFI on test data.

IMPLICATIONS OF PFI

Can we get insight into whether the ...

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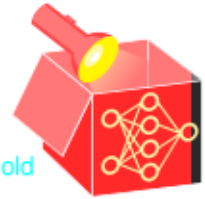
   - $PFI_j \neq 0 \Rightarrow$ model relies on x_j
   - As the training vs. test data example demonstrates, the converse does not hold



Interpretable Machine Learning – 8 / 12

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2. feature x_j contains prediction-relevant information?
   - $PFI_j = 0 \Rightarrow x_j$ is dependent of y or its covariates x_{-j} or both (due to extrapolation)
   - $PFI_j \neq 0 \Rightarrow x_j$ is dependent of y or it's covariates x_{-j} or both (due to extrapolation)
   - x_j is not exploited by model (regardless of whether it is useful for y or not) $\Rightarrow PFI_j = 0$

Interpretable Machine Learning – 8 / 12

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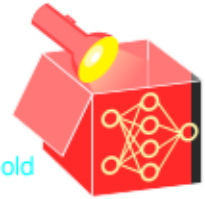

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2. feature x_j contains prediction-relevant information?
   - $PFI_j \neq 0 \Rightarrow x_j$ is dependent of y or it's covariates x_{-j} or both (due to extrapolation)
3. model requires access to x_j to achieve it's prediction performance?
   - x_j is not exploited by model (regardless of whether it is useful for y or not) $\Rightarrow PFI_j = 0$
   - As the extrapolation example demonstrates, such insight is not possible

Interpretable Machine Learning – 8 / 12

TESTING IMPORTANCE (PIMP)

Altmann et al. (2010)

- PIMP was originally introduced for random forest's built-in permutation feature importance

Interpretable Machine Learning – 9 / 12

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Interpretable Machine Learning – 9 / 12

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- Sampling under H_0: Permute target y, retrain model, compute PFI scores (repeat)
  - ⇒ By computing PFI scores again, we obtain distribution of PFI scores under H_0

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  - ⇒ By computing PFI scores again, we obtain distribution of PFI scores under H_0
- Compute p-value - the tail probability under H_0 - and use it as a new importance measure

Interpretable Machine Learning – 9 / 12

TESTING IMPORTANCE (PIMP)

PIMP algorithm:

1. For $m \in \{1, 1, \ldots, n_{repetitions}\}$:
   - Permute response vector y
   - Retrain model with data $\mathbf{X}$ and permuted y
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Interpretable Machine Learning – 10 / 12

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Interpretable Machine Learning – 10 / 12

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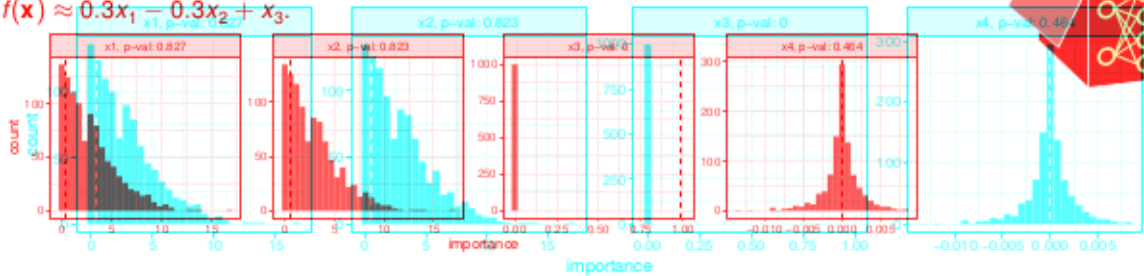
2. Train model with $\mathbf{X}$ and unpermuted y
3. For each feature $j \in \{1, \ldots, p\}$:
   - Fit probability distribution of the feature importance values PFI_j^m, $m \in \{1, \ldots, n_{repetitions}\}$ (choice between Gaussian, lognormal, gamma or non-parametric)
   - Compute feature importance PFI_j for the model without permutation of y (under H_1)
   - Retrieve the p-value of PFI_j based on the fitted distribution

Interpretable Machine Learning – 10 / 12

PIMP FOR EXTRAPOLATION EXAMPLE

Recall: $y = x_3 + \epsilon_y$ with $\epsilon_y \sim N(0, 0.1)$, x_1, x_2 highly correlated but independent of y, x_4 is independent of y and all other variables. Fitting a LM yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$.



- Histograms: H_0 distribution of PFI scores after permuting y (1000 repetitions)
- Red: PFI score estimated on unpermuted y (under H_1) $\rightsquigarrow$ compare against H_0 distribution
- Results: Although PFI for x_1 and x_2 is nonzero (red), PIMP considers them not significantly relevant (p-value > 0.05)

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Interpretable Machine Learning – 12 / 12

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Interpretable Machine Learning – 12 / 12

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Interpretable Machine Learning – 12 / 12

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Interpretable Machine Learning – 12 / 12